

CO-1 OUTPUT PREDICTION

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Course code:CSA0307

Course name: DATA STRUCTURES

QUESTION-1:

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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QUESTIONS

Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

1/3 answered

Q1 Searching 1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n, key, count = 0;
4     printf("enter the n value:\n");
5     scanf("%d", &n);
6     printf("enter the values one by one:\n");
7     int arr[n];
8     for (int i = 0; i < n; i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("enter the key value:\n");
12    scanf("%d", &key);
13    for (int i = 0; i < n; i++) {
14        if (arr[i] == key) {
15            count++;
16        }
17    }
18    printf("%d", count);
19    return 0;
20 }
```

Input

2
3
4
4
4

Output

enter the n value:
enter the values one by one:
enter the key value:
2

  

QUESTION-2:

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QUESTIONS

Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Q2 Array 1 mark

Write a program for binary search

Your Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int target) {
3     int left = 0, right = n - 1;
4     while (left <= right) {
5         int mid = left + (right - left) / 2;
6         if (arr[mid] == target)
7             return mid;
8         else if (arr[mid] < target)
9             left = mid + 1;
10        else
11            right = mid - 1;
12    }
13    return -1;
14 }
15 int main() {
16     int n, target;
17     printf("Enter number of elements: \n");
18     scanf("%d", &n);
```

Input

```
5
1 3 5 7 9
7
```

Output

```
Enter number of elements:
Enter sorted elements:
Enter target element:
Element found at index 3
```

QUESTION-3:

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QUESTIONS

Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Q3 Merging 1 mark

Write a C program for Merge Two Arrays

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, m;
5     printf("Enter size of first array (n): ");
6     scanf("%d", &n);
7
8     int a[n];
9     printf("Enter %d elements (sorted): ", n);
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &a[i]);
12    }
13
14    printf("Enter size of second array (m): ");
15    scanf("%d", &m);
16
17    int b[m];
18    printf("Enter %d elements (sorted): ", m);
```

Input

```
5
1 2 4 3 5
5
1 2 6 7 8
```

Output

```
Enter size of first array (n):
Enter 5 elements (sorted): Enter
size of second array (m): Enter 5
elements (sorted): Merged Array: 1
1 2 2 4 3 5 6 7 8
```